Corollary. $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations.

The following lemma, which will be useful in the next development, is an immediate consequence of Theorem 2.11 and Lemma 2.12 (i).

Lemma 2.14. *Under Models IIa—VIIa, a linear function* $\underline{\lambda}'\underline{\beta}$ *is estimable if and only if* $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix}$.

The equations that play the role of the conjugate normal equations in our development are the constrained conjugate normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix},$$

that is, $X'X\underline{r}_1 + A'\underline{r}_2 = \underline{\lambda}$ and $A\underline{r}_1 = \underline{0}$.

Theorem 2.16. *Under Models IIa—VIIa,* $\underline{\lambda}'\underline{\beta}$ *is estimable, that is,*

$$\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X \\ A \end{bmatrix}, \quad \textit{that is,} \quad \underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix},$$

if and only if the constrained conjugate normal equations are consistent.

Proof. For the "only if" part ($\Longrightarrow$), suppose that $\underline{\lambda}'\underline{\beta}$ is estimable, or $\underline{\lambda}' \in \mathcal{R}\begin{bmatrix} X'X \\ A \end{bmatrix}$, implying $\underline{\lambda}' = \underline{r}_1'X'X + \underline{r}_2'A$ for some $\underline{r}_1$ and $\underline{r}_2$. It suffices to use Lemma 2.10: take $\underline{a} = \begin{bmatrix} \underline{a}_1 \\ \underline{a}_2 \end{bmatrix}$ to be any vector such that

$$\underline{a}' \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1', \underline{a}_2') \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = (\underline{a}_1'X'X + \underline{a}_2'A, \underline{a}_1'A') = \underline{0}'$$

and show

$$\underline{a}' \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix} = 0.$$

Corollary. Under Models IIa—VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}^- \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}.$$

Theorem 2.17. *Under Models IIa—VIIa, the least squares estimate of an estimable function* $\underline{\lambda}'\underline{\beta}$ *is given by*

$$\underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b},$$

where $\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix}$ *is any solution to the constrained conjugate normal equations.*

Proof. For any solution $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ to the constrained normal equations and any solution $\begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix}$ to the constrained conjugate normal equations, we have $X'X\widehat{\underline{\beta}} + A'\underline{\rho} = X'\underline{y}$, $A\widehat{\underline{\beta}} = \underline{b}$, $X'X\underline{r}_1 + A'\underline{r}_2 = \underline{\lambda}$, and $A\underline{r}_1 = \underline{0}$.

Under Models IIa—VIIa, the least squares estimator of an estimable function $\underline{\lambda}'\underline{\beta}$ is an unbiased estimator. To see this, by Theorem 2.17, the least squares estimator $\underline{r}_1'X'\underline{y} + \underline{r}_2'\underline{b}$ is of the form $c + \underline{a}'\underline{y}$, with $c = \underline{r}_2'\underline{b}$ and $\underline{a}' = \underline{r}_1'X'$. That is, note

$$c = \underline{r}_2'\underline{b} \qquad \& \qquad \underline{a}'X + \underline{r}_2'A = \underline{r}_1'X'X + \underline{r}_2'A = \underline{\lambda}'$$

Suppose that the two vectors $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable, and let $\widehat{\underline{\beta}}$ represent any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta})$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$ (i.e., from a solution to constrained normal equations).

Take $\begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$ and $\begin{bmatrix} S_1 \\ S_2 \end{bmatrix}$ to be any solutions to

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} \Lambda_1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} \Lambda_2 \\ 0 \end{bmatrix}.$$

That is, $X'XR_1 + A'R_2 = \Lambda_1$, $AR_1 = 0$, $X'XS_1 + A'S_2 = \Lambda_2$, and $AS_1 = 0$.

Then, by Theorem 2.17, $\Lambda_1'\widehat{\underline{\beta}} = R_1'X'\underline{y} + R_2'\underline{b}$, and $\Lambda_2'\widehat{\underline{\beta}} = S_1'X'\underline{y} + S_2'\underline{b}$.

Under Model IIa, we have

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = R_1'X'(\sigma^2 I)XS_1 = \sigma^2 R_1'X'XS_1$$

$$= \sigma^2 R_1'(\Lambda_2 - A'S_2) = \sigma^2 R_1'\Lambda_2 - \sigma^2(AR_1)'S_2 = \sigma^2 R_1'\Lambda_2,$$

and

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2(X'XR_1)'S_1 = \sigma^2(\Lambda_1 - A'R_2)'S_1 = \sigma^2\Lambda_1'S_1 - \sigma^2 R_2'(AS_1) = \sigma^2\Lambda_1'S_1.$$

It can also be shown (again, under Model IIa) that, for any generalized inverse $\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$,

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'G_{11}\Lambda_2.$$

The proof of the following parallels that of the ordinary Gauss-Markov theorem.

Theorem 2.18. *Under Models IIa—VIIa, if* $\underline{\lambda}'\underline{\beta}$ *is estimable and* $\widehat{\underline{\beta}}$ *is the first component of any solution to the constrained normal equations, then* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is a linear unbiased estimator of* $\underline{\lambda}'\underline{\beta}$*. Under Model IIa, the variance of* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is uniformly less than that of any other linear unbiased estimator of* $\underline{\lambda}'\underline{\beta}$*, that is,* $\underline{\lambda}'\widehat{\underline{\beta}}$ *is the best (minimum variance) linear unbiased estimator (BLUE) of* $\underline{\lambda}'\underline{\beta}$*.*

We can also find an unbiased estimator of σ^2 under Model IIa:
Take $\begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix}$ to be any solution to the constrained normal equations, and take $\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ to be any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$.

For the residual sum of squares, we have

$$Q(\widehat{\underline{\beta}}) = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}}) = \underline{y}'(\underline{y} - X\widehat{\underline{\beta}}) - \widehat{\underline{\beta}}'(X'\underline{y} - X'X\widehat{\underline{\beta}})$$
$$= (\underline{y} - X\widehat{\underline{\beta}})'\underline{y} - \widehat{\underline{\beta}}'A'\underline{\rho} = (\underline{y} - X\widehat{\underline{\beta}})'\underline{y} - \underline{b}'\underline{\rho} = \underline{y}'\underline{y} - \widehat{\underline{\beta}}'X'\underline{y} - \underline{\rho}'\underline{b}$$
$$= \underline{y}'\underline{y} - (G_{11}X'\underline{y} + G_{12}\underline{b})'X'\underline{y} - (G_{21}X'\underline{y} + G_{22}\underline{b})'\underline{b}.$$

It can be shown that, under Model IIa,

$$E[Q(\widehat{\underline{\beta}})] = \sigma^2 \left\{ n - \text{rank} \begin{bmatrix} X \\ A \end{bmatrix} + \text{rank}(A) \right\}.$$

Thus, under Model IIa,

$$\widehat{\sigma}^2 = Q(\widehat{\underline{\beta}})/(n-f) = (\underline{y}'\underline{y} - \widehat{\underline{\beta}}'X'\underline{y} - \underline{\rho}'\underline{b})/(n-f)$$

is an unbiased estimator of σ^2, where $f = \text{rank} \begin{bmatrix} X \\ A \end{bmatrix} - \text{rank}(A)$.

Instead of imposing equality constraints $A\underline{\beta} = \underline{b}$ on the parameter vector $\underline{\beta}$, we might wish to impose q inequality constraints, say

$$A\underline{\beta} \geq \underline{b}.$$

Here, the least squares problem is that of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$ subject to the constraints $A\underline{\beta} \geq \underline{b}$ or, equivalently, that of minimizing $Q(\underline{\beta})$ with respect to $\underline{\beta}$ and $\underline{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_q \end{bmatrix}$ subject to the constraints

$$\begin{aligned} A\underline{\beta} &= \underline{b} + \underline{u}, \\ u_i &\geq 0, \quad i = 1, \ldots, q. \end{aligned}$$

The quantities $u_1, \ldots, u_q$ are called surplus variables.

By taking advantage of our results on the least squares problem with equality constraints, this problem can be reduced essentially to that of minimizing the convex function

$$\left[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\right]' \left[\underline{y} - XG_{11}X'\underline{y} - XG_{12}(\underline{b} + \underline{u})\right]$$

with respect to $\underline{u}$ subject to the linear equality constraints $(I - AA^{-})(\underline{b} + \underline{u}) = \underline{0}$ and the inequality constraints $u_i \geq 0, i = 1, \ldots, q$. This is a quadratic programming problem.

2.10 Extensions to Model III

First, we have the following lemma.

Lemma 2.15. *Corresponding to any $n \times n$ nonnegative definite matrix A of rank $r \geq 1$, there exists an $r \times n$ matrix S such that*

$$A = S'S.$$

Moreover, for any $r \times n$ matrix S such that $A = S'S$, $\text{rank}(S) = r$.

Proof. Since A is a symmetric matrix, there exists a matrix Q such that $Q'AQ = D$, where D is a diagonal matrix with the eigenvalues of A along the main diagonal, and where Q is an orthogonal matrix, that is, $Q'Q = QQ' = I$. (See, for example, Section 11.5 and Chapter 11A in Searle's book *Matrix Algebra Useful for Statistics*.) Because A is nonnegative definite, the diagonal entries of D must be non-negative and the number of positive entries must match the rank of A.

Since $\text{rank}(A) = r$, we may take $D = \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix}$, where D_r is an $r \times r$ nonsingular matrix, with all its diagonal elements being positive. Thus, we can define $D^{1/2}$ as the $r \times r$ nonsingular diagonal matrix whose diagonal elements are the square roots of those of D_r. Then,

$$A = QDQ' = Q \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} Q' = Q \begin{bmatrix} D_r^{1/2} \\ 0 \end{bmatrix} \begin{bmatrix} D_r^{1/2} & 0 \end{bmatrix} Q' = S'S,$$

for an $r \times n$ matrix $S = \begin{bmatrix} D_r^{1/2} & 0 \end{bmatrix} Q'$.

Moreover, if S is any $r \times n$ matrix such that $A = S'S$, then by Lemma 2.3,

$$\text{rank}(S) = \text{rank}(S'S) = \text{rank}(A) = r.$$

□

In Model III, $\text{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given nonnegative definite matrix.

Assume (first) that H is nonsingular to develop extensions to previous results.

Write $H = R'R$ for an $n \times n$ nonsingular matrix R, where $H^{-1} = R^{-1}(R^{-1})'$.
Note that

$$\begin{aligned} \underline{y} = X\underline{\beta} + \underline{\epsilon} \quad &\Longleftrightarrow \quad (R^{-1})'\underline{y} = (R^{-1})'X\underline{\beta} + (R^{-1})'\underline{\epsilon} \\ &\Longleftrightarrow \quad \underline{z} = W\underline{\beta} + \underline{h}, \end{aligned}$$

where

$$\underline{z} = (R^{-1})'\underline{y}, \qquad W = (R^{-1})'X, \qquad \underline{h} = (R^{-1})'\underline{\epsilon}$$

and

$$E[\underline{h}] = E[(R^{-1})'\underline{\epsilon}] = (R^{-1})'E(\underline{\epsilon}) = \underline{0}$$

with

$$\text{Var}[\underline{h}] = \text{Var}[(R^{-1})'\underline{\epsilon}] = (R^{-1})'(\sigma^2 H)R^{-1} = \sigma^2(R^{-1})'R'RR^{-1} = \sigma^2 I.$$

We will refer to the model

$$\underline{z} = W\underline{\beta} + \underline{h}$$

as the "transformed model," where the distribution of $\underline{h}$ is assumed to be identical to that of $(R^{-1})'\underline{\epsilon}$ under (the original) Model III and where the parameter space is taken to be $\left\{\underline{\beta}, \sigma^2 : \underline{\beta} \in \mathbb{R}^p, \sigma^2 > 0\right\}$.
The transformed model is of the same general form as Model II, with $E(\underline{h}) = \underline{0}$ and $\text{Var}(\underline{h}) = \sigma^2 I$.

Because $\underline{h}$ and $(R^{-1})'\underline{\epsilon}$ have the same distribution, the distribution of $\underline{z}$ or $\underline{y}$, as derived under the transformed model, is identical to the distribution of $\underline{z}$ or $\underline{y}$, as derived under Model III.

Note that $\mathcal{R}(W) = \mathcal{R}[(R^{-1})'X] = \mathcal{R}(X)$. Then, by Theorem 2.1,

a linear function $\underline{\lambda}'\underline{\beta}$ is estimable under the transformed model

$\iff$ $\underline{\lambda}' \in \mathcal{R}(W)$

$\iff$ $\underline{\lambda}' \in \mathcal{R}(X)$

$\iff$ $\underline{\lambda}'\underline{\beta}$ is estimable under Model III.

Furthermore, by Lemma 2.1, for any estimable function $\underline{\lambda}'\underline{\beta}$,

$c + \underline{a}'\underline{z}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ under the transformed model

$\iff$ $c = 0$ and $\underline{\lambda}' = \underline{a}'W$

$\iff$ $c = 0$ and $\underline{\lambda}' = \underline{a}'(R^{-1})'X$

$\iff$ $c + \underline{a}'(R^{-1})'\underline{y}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ under Model III

$\iff$ $c + \underline{a}'\underline{z}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$ under Model III.

Thus, the transformed model and Model III have the same set of estimable functions and the same set of linear unbiased estimators for every estimable $\underline{\lambda}'\underline{\beta}$.

Take $\widehat{\underline{\beta}}$ to be any solution to the equations

$$W'W\widehat{\underline{\beta}} = W'\underline{z}, \qquad \text{or equivalently,} \quad X'H^{-1}X\widehat{\underline{\beta}} = X'H^{-1}\underline{y},$$

which are known as the Aitken equations. Here,

$$W'W = \left[(R^{-1})'X\right]'(R^{-1})'X = X'R^{-1}(R^{-1})'X = X'(R'R)^{-1}X = X'H^{-1}X$$

and

$$W'\underline{z} = \left[(R^{-1})'X\right]'(R^{-1})'\underline{y} = X'R^{-1}(R^{-1})'\underline{y} = X'(R'R)^{-1}\underline{y} = X'H^{-1}\underline{y}.$$

By the Gauss-Markov theorem, under the transformed model, $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of an estimable function $\underline{\lambda}'\underline{\beta}$, and the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator. Thus, we have the following.

Theorem 2.19. *Under Model III, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the Aitken equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, and the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator.*

Note that $\widehat{\underline{\beta}}$ is a solution to the Aitken equations if and only if $(\underline{z} - W\underline{\beta})'(\underline{z} - W\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, or equivalently if and only if $(\underline{y} - X\underline{\beta})'H^{-1}(\underline{y} - X\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$. The problem of minimizing

$$(\underline{y} - X\underline{\beta})'A(\underline{y} - X\underline{\beta})$$

with respect to $\underline{\beta}$ is called generalized least squares, where A is a positive definite matrix.

If $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable vectors and if $\widehat{\underline{\beta}}$ is a solution to the Aitken equations, then

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'(W'W)^-\Lambda_2 = \sigma^2\Lambda_1'(X'H^{-1}X)^-\Lambda_2.$$

Also, $p^* = \text{rank}(X) = \text{rank}(W)$. Thus, an unbiased estimator of σ^2 is

$$\widehat{\sigma}^2 = (\underline{z} - W\widehat{\underline{\beta}})'(\underline{z} - W\widehat{\underline{\beta}})/(n - p^*) = (\underline{y} - X\widehat{\underline{\beta}})'H^{-1}(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*).$$